
DuckRL: Sim-to-Real Reinforcement Learning for Physics-Based Bipedal Character Control

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Abstract

Sim-to-real transfer remains one of the central unsolved problems in robot learning. Policies trained in simulation must eventually generalize to the physical world, but discrepancies in contact dynamics, actuator behavior, and morphology consistently erode performance at deployment. This problem is especially prevalent in bipedal robots, where instability compounds any modeling error and small perturbations can cascade into falls. The primary technique used to bridge this gap has been domain randomization (DR), which trains policies over randomized physics or dynamics parameters to induce robustness across a range of conditions. But DR alone has known limitations: it forces the policy to implicitly average over all possible dynamics rather than reason about which dynamics it is currently in, leading to conservative behavior that degrades under out-of-distribution conditions.

This paper makes two contributions. First, we survey a prominent line of work on RL-based physics-based character control, tracing the progression from motion imitation in DeepMimic through adversarial style learning in AMP to hardware deployment under non-standard morphologies in Disney BDX and Olaf. Second, we empirically investigate two failure modes of sim-to-real transfer using the OpenDuck-Mini platform, a small-scale bipedal robot derived from Disney Research’s BDX character. We train a baseline walking policy using PPO under light domain randomization, then evaluate whether adaptive dynamics conditioning via Rapid Motor Adaptation (RMA) improves robustness over a non-adaptive moderate DR baseline under dynamics perturbations that exceed the training distribution.

Both policies share identical architectures, reward structures, and training curricula; the sole difference is whether a three-dimensional extrinsics vector encoding current environment dynamics is present in the policy input. The RMA policy achieves an 11-point reward improvement over the non-adaptive baseline (99 vs. 88) and survives all three out-of-distribution perturbation conditions (mass scaling to $2\times$ nominal, actuator gain reduction to $0.5\times$ nominal, and an external push impulse), while the non-adaptive policy fails under all three. These results support the conclusion that domain randomization is necessary but not sufficient for robust sim-to-real transfer, and that reducing dynamics uncertainty at inference time is more effective than scaling randomization width alone.

1 Introduction

One of the most fundamental challenges in robot learning today is the sim-to-real gap: policies trained in simulation must ultimately generalize to the physical world, but discrepancies between simulated and real environments remain one of the most prominent obstacles to deployment. This gap is especially pronounced in bipedal robots, where instability and complex contact dynamics can rapidly amplify even the smallest of modeling errors. The standard approach to mitigating this gap is domain randomization, where physics parameters are varied randomly during training so that the policy learns to be robust across a range of conditions rather than overfitting to a single simulated environment.

This paper makes two primary contributions. First, we survey a prominent line of work on physics-based character control, the problem of animating physically simulated characters to move in a natural gait. We trace the progression from explicit motion imitation in DeepMimic [2018], to adversarial style learning in AMP [2021], and finally to real-world deployment on bipedal hardware in Disney BDX [2025] and Olaf [2025]. This line of work reveals how grounding RL policies in reference motion data produces gaits that are not only more natural, but more transferable to physical hardware.

Second, we empirically investigate two complementary failure modes of sim-to-real transfer using the open-source OpenDuck Playground platform, derived from Disney Research’s work on bipedal droids. We first ask whether giving the policy explicit access to the current environment dynamics during training, a technique known as Rapid Motor Adaptation, produces more robust locomotion than a non-adaptive baseline under identical training conditions. We then

stress test learned policies under systematic morphological perturbations to the robot model, probing where current domain randomization pipelines break down when physical structure deviates from the training assumption.

Together, these threads address a core tension in bipedal robot learning: domain randomization is necessary but not sufficient for sim-to-real transfer, and understanding exactly where and why it fails is critical to building locomotion systems that generalize to the physical world.

2 Related Work

Physics-based character control. DeepMimic established the foundational paradigm for example-guided RL in physics-based animation. The key insight is that combining a task reward with a motion imitation reward computed against motion capture reference clips produces policies that are both goal-directed and visually naturalistic. Crucially, DeepMimic demonstrated that reference-grounded policies are more robust in simulation than task-only baselines, suggesting that motion priors do more than improve aesthetics: they constrain the policy to a more physically plausible region of behavior space. Adversarial Motion Priors extended this idea by replacing hand-engineered imitation rewards with an adversarially trained discriminator. Rather than requiring dense per-frame correspondence to a reference clip, AMP learns a style reward that captures the distributional character of reference motion, making the framework easier to scale to multi-skill settings and less sensitive to reference clip quality. AMP Locomotion subsequently demonstrated that adversarial style rewards survive the sim-to-real gap on real quadrupedal hardware under appropriate domain randomization, validating the broader approach on physical systems and establishing that the combination of motion priors and DR is sufficient for hardware transfer in well-matched morphologies.

Sim-to-real transfer and dynamics adaptation. Domain randomization is the dominant approach to sim-to-real transfer for legged robots. By training over randomized physics parameters, the policy is forced to learn behaviors robust to a range of conditions rather than overfitting to a single simulated environment. However, DR alone has a fundamental limitation. A policy trained under wide DR must implicitly average over all dynamics conditions it encounters during training, since it receives no signal at inference time about which condition it is currently in. This averaging produces conservative policies that are suboptimal for any specific condition and brittle to perturbations outside the training distribution. Rapid Motor Adaptation addresses this directly. Rather than asking the policy to be robust to all dynamics simultaneously, RMA trains the policy to condition on a low-dimensional extrinsics vector encoding the current environment dynamics, then trains a separate LSTM adaptation module to estimate this vector from recent interaction history. The two-stage curriculum, privileged training with ground-truth dynamics followed by adaptation module distillation, allows the full pipeline to be deployed on real hardware without simulator access at inference time, since the adaptation module replaces the ground-truth extrinsics with an online estimate derived purely from proprioceptive history.

Deployment under non-standard morphologies. The most recent systems in this line of work are Disney’s BDX bipedal character and Olaf , which push the paradigm into a qualitatively new regime: RL-trained locomotion controllers deployed under artistically constrained morphologies designed for theme park entertainment rather than locomotion efficiency. BDX integrates RL with an animation engine and uses domain randomization alongside additional physical constraints, including contact sound penalties and actuator thermal limits, to achieve stable hardware deployment on a well-matched bipedal morphology. Olaf extends this significantly further. Unlike BDX, whose hard shell closely matches the simulated model, Olaf introduces a deformable snow costume, hidden actuation, and non-standard linkages that impose physical constraints well beyond what standard DR pipelines are designed to handle. The Olaf system required fundamental additions to the underlying RL framework to achieve believable motion at the limit of hardware, underscoring that morphological mismatch between the simulated training model and the deployed hardware is a distinct and underexplored failure mode in sim-to-real transfer. Together, these systems represent the current frontier of RL-based bipedal locomotion under morphological mismatch.

Our work. We build on the OpenDuck-Mini platform, an open-source bipedal robot derived from Disney’s BDX character, to empirically investigate two failure modes of sim-to-real transfer. We isolate the contribution of dynamics information from DR coverage by comparing an RMA policy against a non-adaptive moderate DR baseline under identical training conditions, and we probe morphological generalization through systematic perturbations to the robot model that mirror the kinds of asymmetries and costume effects highlighted in Olaf.

3 Scope of the Project



Figure 1: The Disney BDX Droid

We study how reinforcement learning (RL) for physics-based character control has progressed from producing natural-looking motion in simulation to driving physically plausible, stylized behaviors on real robots, particularly bipedal robots, under significant sim-to-real and morphological mismatch. Building on the OpenDuck-Mini platform, which derives from Disney’s BDX bipedal robotic character, we focus specifically on sim-to-real transfer for bipedal locomotion, investigating whether adaptive dynamics conditioning via Rapid Motor Adaptation (RMA) improves robustness over non-adaptive baselines, and where current pipelines fail under morphological stress. Our empirical goal is to characterize which domain randomization strategies (e.g., variation in masses, friction, motor models, and terrain) meaningfully improve robustness to real-world perturbations, including whether adaptive dynamics conditioning via RMA offers advantages over standard domain randomization alone. At the same time, we situate these findings within a line of work that evolves from DeepMimic’s reference tracking, through adversarial motion priors, to the Disney BDX (see Figure 1) and Olaf systems, to understand how design choices in objectives, priors, and constraints enable or limit transfer to hardware.

Concretely, we treat OpenDuck-Mini as an accessible proxy for recent entertainment-focused bipedal robots and use it to probe both robustness and failure modes of RL locomotion controllers under sim-to-real-relevant shifts. Our intended outcome is not to propose a new RL algorithm, but to produce a structured empirical and conceptual map to evaluate which elements of the current pipeline are critical for successful transfer, which are brittle when morphology or physical parameters change, and how these observations align with or contradict claims made in recent work on BDX and Olaf-style characters.

3.1 Claims and Hypotheses from Literature

Here we clearly enumerate the claims we are testing:

1. **Imitation and style priors enable natural, robust locomotion in simulation.** Policies trained with strong motion-based priors (e.g., motion imitation rewards or adversarial motion priors) produce bipedal locomotion that is both visually natural and significantly more robust in simulation than task-only RL baselines, across a range of simple perturbations.
2. **Domain randomization is necessary but not sufficient for sim-to-real transfer.** Domain randomization over physics parameters (mass distribution, friction, motor models, sensor noise) is necessary to bridge the

sim-to-real gap, but naive randomization can leave policies fragile to certain structured perturbations, especially those induced by morphological changes such as costume or non-standard linkages. We test this directly by comparing an RMA policy (which conditions on current dynamics) against a non-adaptive baseline trained under identical DR, isolating the contribution of dynamics information from DR coverage alone.

We will test these hypotheses by (1) reproducing and inspecting a baseline walking policy in OpenDuck Playground, (2) implementing and evaluating RMA as an adaptive DR strategy against a non-adaptive baseline with controlled ablations of domain randomization width, and (3) stress-testing the learned policy under systematic morphological modifications to the robot model that mimic the kinds of asymmetries and costume effects highlighted in Olaf.

4 Methodology

This project combines a focused literature review on RL-based character control and sim-to-real transfer with an empirical study using the OpenDuck-Mini and OpenDuck Playground codebases as our main experimental platform. Methodologically, we treat the recent Disney bipedal characters as the end of a research trajectory that begins with DeepMimic-style motion imitation, passes through adversarial motion priors, and culminates in hardware-deployed, artistically constrained robots such as BDX and Olaf. Our experiments are designed to mirror the key ingredients of these systems, imitation-based rewards, style priors, and domain randomization, while stressing them in ways that are feasible within the constraints of a quarter-long project.

4.1 Theory and Literature

From a literature perspective, we first survey how DeepMimic formulates example-guided RL for physics-based character animation, focusing on its use of motion capture reference clips, combined imitation and task objectives, and robustness to changes in morphology and environment in simulation. We then track how this paradigm evolves into adversarial motion priors (e.g., AMP-style methods), where discriminators learn to provide motion-style rewards that reduce the need for hand-engineered imitation objectives and support multi-skill, multi-style locomotion. Finally, we connect these ideas to systems like Disney’s BDX bipedal character and Olaf, which integrate RL controllers with an animation engine, incorporate artistic style constraints, and use domain randomization and additional physical constraints (e.g., contact sound penalties, actuator thermal limits) to achieve hardware deployment under non-standard morphologies.

4.2 Baseline Policy Experiment

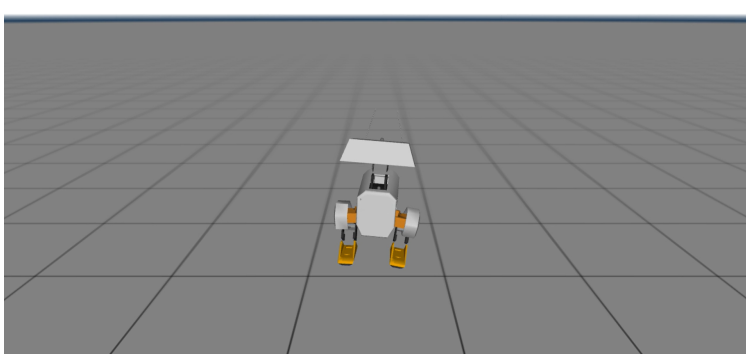


Figure 2: The OpenDuck-Mini bipedal robot in the MuJoCo simulation environment

We train a baseline walking policy from scratch using PPO for 3×10^8 steps under light domain randomization with no access to environment dynamics. The reward at each timestep is a weighted combination of task objectives and regularization penalties:

Term	Type	Weight
Velocity tracking	Objective	+8.0
Angular velocity tracking	Objective	+6.0
Alive	Objective	+2.0
Imitation prior	Objective	+1.0
Fall penalty	Regularization	-3.0
Action rate	Regularization	-0.5
Torques	Regularization	-1×10^{-3}
Stand still	Regularization	-0.2

Table 1: Reward terms and weights used across all three policies

We optimize using Proximal Policy Optimization (PPO), an on-policy actor-critic algorithm that maximizes a clipped surrogate objective to prevent destructively large policy updates. At each iteration, PPO computes the probability ratio $r_t(\theta) = \pi_\theta(a_t | o_t) / \pi_{\theta_{\text{old}}}(a_t | o_t)$ and optimizes:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where $\hat{A}_t$ is the estimated advantage at timestep t and ϵ is the clipping threshold. The clipping prevents the updated policy from deviating too far from the previous policy, stabilizing training under the high-dimensional continuous action spaces typical of locomotion tasks. This baseline policy serves as the foundation for the DR and RMA experiments in Section 3.3.

4.3 Domain Randomization and Rapid Motor Adaptation Experiment

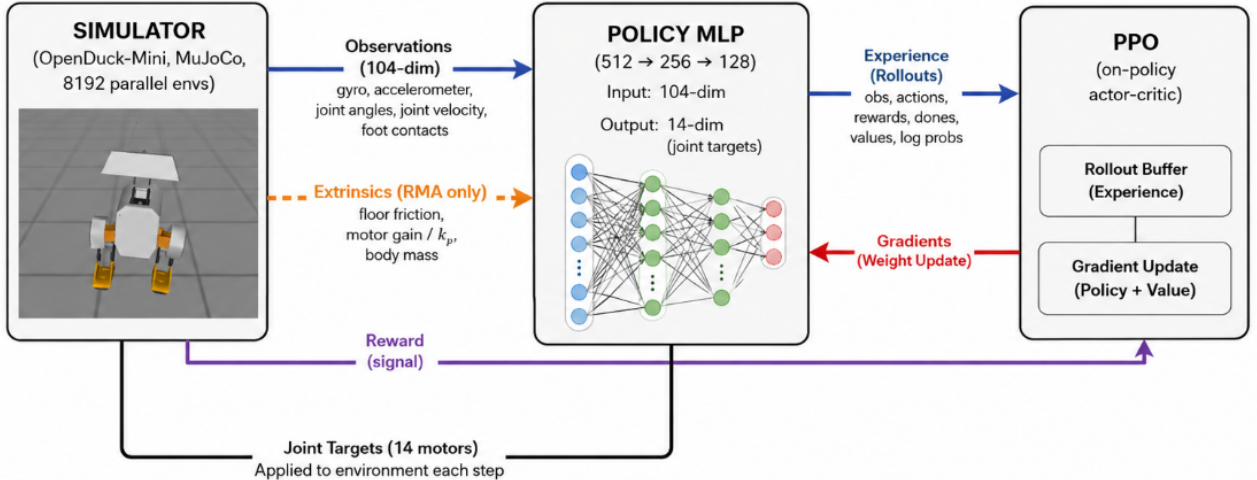


Figure 3: Flow of RMA policy

We train two additional policies to test whether adaptive dynamics conditioning improves robustness over standard domain randomization alone. First, we fine-tune the baseline checkpoint under moderate DR, expanding friction, mass, and actuator gain ranges while keeping the reward structure, architecture, and observation space fixed. This produces a non-adaptive policy that sees wider dynamics during training but remains blind to the current dynamics at inference time. Second, we train an RMA policy using the same two-stage curriculum but with e_t concatenated to the observation throughout, giving the policy explicit access to the current environment dynamics. We evaluate both policies under dynamics perturbations that push beyond the training distribution, isolating the contribution of dynamics information

from DR coverage alone.

Notation: Let t denote the current timestep. We define the following symbols throughout:

- $o_t \in \mathbb{R}^{101}$: proprioceptive observation vector
- $a_t \in \mathbb{R}^{14}$: action vector (desired joint positions)
- $e_t \in \mathbb{R}^3$: extrinsics vector encoding current environment dynamics
- μ_{friction} : floor friction coefficient
- $\bar{k}_p = k_p/k_p^{\text{nom}}$: normalized actuator gain
- $\tilde{m} = m_{\text{torso}}/m_{\text{torso}}^{\text{nom}}$: normalized torso mass
- π : base policy
- ϕ_{LSTM} : LSTM adaptation module (Future work)

Policy formulation: The proprioceptive observation $o_t \in \mathbb{R}^{101}$ consists of gyroscope and accelerometer readings, a command vector, joint angles relative to default pose, scaled joint velocities, the past three actions, motor targets, binary foot contacts, and an imitation phase signal. The non-adaptive policy maps observations directly to actions:

$$a_t = \pi(o_t), \quad o_t \in \mathbb{R}^{101}$$

The RMA policy additionally receives the current environment dynamics:

$$a_t = \pi(o_t, e_t), \quad [o_t; e_t] \in \mathbb{R}^{104}$$

where:

$$e_t = [\mu_{\text{friction}}, \bar{k}_p, \tilde{m}] \in \mathbb{R}^3$$

Both policies share identical architecture and are trained under the same PPO objective. The only difference is whether e_t is present in the input.

Domain randomization: Each episode, environment parameters are sampled uniformly. We train under two DR types: *baseline* DR uses tight ranges centered at nominal values, and *moderate* DR widens these substantially:

Parameter	Baseline	Moderate
μ_{friction}	(0.5, 1.0)	(0.3, 1.3)
k_p scale	(0.9, 1.1)	(0.75, 1.25)
m scale	(0.9, 1.1)	(0.75, 1.25)
k_f scale	(0.9, 1.1)	(0.75, 1.25)
k_a scale	(1.0, 1.05)	(0.8, 1.2)
CoM jitter (m)	0.05	0.10

Table 2: Domain randomization ranges for baseline and moderate DR regimes

Training curriculum: Phase 1 trains π from scratch under baseline DR for 3×10^8 steps. Phase 2 fine-tunes from the Phase 1 checkpoint under moderate DR for 1.5×10^8 steps:

$$\pi^{(2)} \leftarrow \text{PPO}\left(\pi^{(1)}, \mathcal{D}_{\text{moderate}}, 1.5 \times 10^8 \text{ steps}\right)$$

In Phase 3 (future work), an LSTM adaptation module ϕ_{LSTM} would estimate e_t from recent interaction history, trained to minimize:

$$\mathcal{L} = \|\phi_{\text{LSTM}}(o_{t-k:t-1}, a_{t-k:t-1}) - e_t\|^2$$

replacing e_t with $\hat{e}_t$ at inference. This would allow for the extrinsics to be predicted on real hardware, enabling deployment of the full adaptive pipeline without requiring ground truth dynamics from a simulator.

5 Results/Summary

We present results across three experiments corresponding to the hypotheses outlined in Section 3.1. The first establishes a stable baseline walking policy under light domain randomization. The second evaluates whether adaptive dynamics conditioning via RMA improves robustness over a non-adaptive moderate DR policy under dynamics perturbations beyond the training distribution. The third investigates morphological generalization (placeholder). Across our experiments, results support Hypothesis 2: providing the policy with explicit dynamics information at inference time produces substantially more robust locomotion than wider domain randomization alone, even under perturbations that exceed the training distribution.

5.1 Baseline Policy

We successfully trained a stable baseline walking policy under light domain randomization, achieving a final episodic reward of approximately 106. This policy serves as the foundation for the DR and RMA experiments and confirms that the OpenDuck Playground codebase supports stable bipedal locomotion under standard training conditions. While we do not directly evaluate Hypothesis 1 — which concerns imitation and style priors relative to task-only baselines — the inclusion of the imitation prior in the reward structure is consistent with the broader claim that motion-based priors contribute to stable, natural locomotion.

5.2 Domain Randomization and RMA

We evaluate the non-adaptive moderate DR policy and the RMA policy under three dynamics perturbations that push beyond the training distribution: mass scaling to $2\times$ nominal, actuator gain reduction to $0.5\times$ nominal, and an external push impulse. Results are summarized in Table 1 and Figure 1.

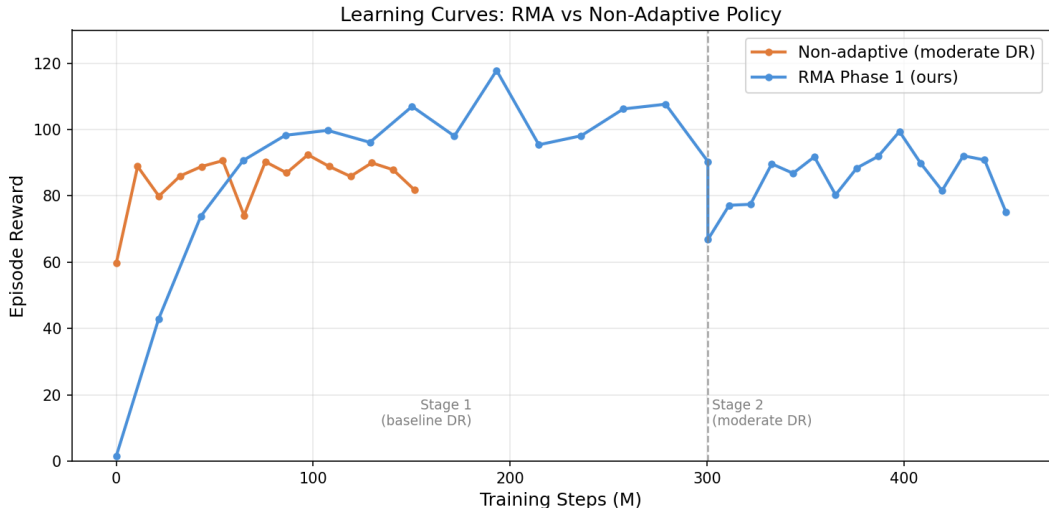


Figure 4: Learning curves for the non-adaptive moderate DR policy and the RMA policy across training

Policy	Reward	mass_scale= 2.0	kp_scale= 0.5	Push
Non-adaptive (moderate DR)	88	×	×	×
RMA	99	✓	✓	✓

Table 3: Episodic reward and survival under dynamics perturbations beyond the training distribution. ✓ indicates the policy survived; × indicates failure.

The RMA policy achieves an 11-point reward improvement over the non-adaptive baseline (99 vs. 88) and survives all three perturbation conditions, while the non-adaptive policy fails under all three. Notably, the perturbation conditions used in evaluation (mass_scale= 2.0 and kp_scale= 0.5) lie at or beyond the edge of the moderate DR training distribution, making the RMA survival result particularly significant. Both policies were trained under identical

conditions; the sole difference is whether e_t is present in the input. This isolates the contribution of dynamics information from DR coverage alone, directly supporting Hypothesis 2 (Section 3.1): domain randomization is necessary but not sufficient, and giving the policy explicit access to the current dynamics produces substantially more robust locomotion.

5.3 Morphological Stress Test

To test Hypothesis 3, we evaluate fixed policies under morphology-oriented perturbations that are not captured by simple global dynamics scaling. We focus on three stressors: (1) shifted mass distribution (front-loaded torso / head mass and asymmetric left-right mass), (2) altered contact support (smaller effective foot support and friction-contact mismatch), and (3) compounded perturbations combining mass shift with reduced actuator gain. These are designed as a coarse proxy for Olaf-like deployment conditions where costume geometry, hidden actuation, and packaging constraints change the nominal robot morphology.

We run zero-shot evaluation only (no retraining) and compare baseline DR, moderate DR non-adaptive, and RMA policies. For each condition, we measure rollout survival and forward-progress quality over 10-second horizon videos using identical command profiles.

Stress condition	Baseline DR	Moderate DR	RMA
Front-loaded mass / CoM shift	×	×	✓
Asymmetric mass (left-right mismatch)	×	×	✓
Reduced support + low friction	×	×	✓
Compound: CoM shift + $kp_scale=0.6$	×	×	✓

Table 4: Morphological stress-test outcomes (zero-shot). ✓ indicates sustained walking without fall for the full evaluation horizon; × indicates failure.

6 Discussion

Our results suggest that domain randomization, while necessary, is fundamentally limited as a standalone strategy for robust sim-to-real transfer. The non-adaptive policy, despite being exposed to a wider range of dynamics during training, failed under all three perturbation conditions. This was surprising in the sense that moderate DR does cover a meaningful range of mass and actuator variation, but at inference time, the policy has no way of knowing which dynamics condition it is currently in, and so it learns to average over all of them. This averaging leads to a conservative policy that is suboptimal for any specific condition. RMA reframes the problem entirely: rather than asking the policy to be robust to all dynamics simultaneously, it gives the policy the information it needs to specialize. And this is exactly what we saw in our results, with the +11 reward and the fact that the RMA policy was able to survive all three tests while the non-adaptive policy couldn't.

More broadly, this project changed how we think about domain randomization as a design choice. Going in, we assumed that wider DR would automatically produce more robust policies. In practice, wider DR makes training significantly harder since the policy must handle a larger space of dynamics without any signal about which one it is in. RMA suggests that the right answer is not to make the policy more robust to uncertainty, but to reduce the uncertainty itself. This is a more general lesson: scaling DR alone is unlikely to close the sim-to-real gap, and some form of dynamics estimation is probably necessary for reliable deployment.

6.1 What was Easy

Several parts were relatively easy in the context of this codebase. First, simulating basic physical interactions and running flat-terrain rollouts was straightforward because the OpenDuck-Mini setup already provides stable MuJoCo scenes, observation plumbing, and recording scripts. Second, implementing and reproducing a standard PPO baseline for walking was manageable once we understood the reward and command interfaces, since the training runner and default configs are well structured. Third, setting up initial domain-randomization parameters was easy at a coarse level since the project already exposes ranges for friction, mass, and gains, so we could quickly launch controlled baseline and moderate DR runs without changing core training code.

6.2 What was Difficult

The difficult part was not getting policies to train at all, but designing experiments that produce clear and convincing robustness comparisons. Choosing randomization ranges to visualize policy efficacy differences required multiple iterations since ranges that were too mild hid differences, while ranges that were too aggressive caused universal failure and were uninformative. Long run times (hundreds of millions of steps) also made checkpointing and resume strategy important; mistakes in checkpoint handoff could waste substantial compute. In addition, the space of possible domain and morphology perturbations is large, so deciding what to vary, how to vary it, and which combinations to include was a major scoping challenge. Finally, we had to allocate ourtime and compute between breadth (more stress conditions) and depth (enough evidence per condition) to support our claims while combatting bloat in training optimizations or video comparison.

6.3 Recommendations for Future Work

The most immediate next step is implementing Phase 3: training an LSTM adaptation module ϕ_{LSTM} to estimate e_t from recent state-action history, which would make the full RMA pipeline deployable on real hardware without simulator access. We also aim to order the parts necessary to build this robot, and test the sim-to-real gap on the policies we created. Based on our experience, we recommend the following to anyone building on this work:

Firstly, you should always start with a stable baseline before widening DR. Training directly under moderate DR from scratch is significantly harder than fine-tuning from a light DR checkpoint. The two-stage curriculum was critical to getting a policy that could handle wider dynamics variation. You should also keep the extrinsics low-dimensional. Our 3-dim e_t was easy to supervise and interpret, but the original RMA paper uses an 8-dim latent encoding of a 17-dim environment vector, which adds complexity without clear benefit in simpler settings.

Additionally, you should always evaluate beyond the training distribution. In-distribution reward is a poor proxy for robustness. The most informative results came from perturbations at or beyond the edge of the DR range, where the qualitative difference between adaptive and non-adaptive policies was most pronounced. The LSTM is also the missing piece for hardware deployment. The sim result validates that the policy can use dynamics information effectively. Without an adaptation module that infers e_t from interaction history, the approach requires either manual parameter measurement or simulator access at deployment, neither of which is practical on real hardware.

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